

Product Line Profitability & Margin Performance Analysis.

Nassau Candy Distributors

**Business Analyst Research Paper
Prepared using Python Pandas and Streamlit**

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Abstract

This study examines product-line profitability and margin performance using a business analytics framework developed for Nassau Candy Distributor .The project converts transaction-level information into management-oriented measures of sales, cost, gross profit, profit per unit, risk exposure, profit concentration and margin volatility. An interactive Streamlit dashboard was developed to support product-level and division-level analysis through filters, leaderboards, diagnostic charts, risk flags, Pareto analysis and Management recommendations.

In the dashboard view documented in this study, the analyzed portfolio generated approximately \$141,784 in revenue and \$93,443 in gross profit, corresponding to a 65.9% gross margin and 34.1% cost share of sales. Chocolate was the dominant division by sales and gross profit, while the other division recorded a lower gross margin but higher profit per unit. Profit concentration was substantial: approximately four of the fifteen reviewed products accounted for 80% of cumulative gross profit, indicating meaningful portfolio dependency. One product, Kazookles, was classified as High Risk, with a 7.7% gross margin and 92.3% cost share of sales. Three additional products were classified as Cost Risk and assigned cost-renegotiation recommendations.

The study concludes that product profitability should be managed using a combination of margin, cost exposure, contribution and volatility measures rather than revenue alone. The Streamlit dashboard provides a practical decision-support mechanism for prioritizing corrective actions and monitoring profitability over time.

Keywords: product profitability; gross margin; business intelligence; dashboard; Pareto analysis; cost risk; Streamlit; data-driven decision making

1.Introduction

Organizations operating broad product portfolios face recurring management challenges: sales performance does not necessarily translate into equivalent profitability. Products may differ substantially in cost structure, unit economics, contribution to total profit and volatility. Consequently, management decisions based only on revenue can overlook products that consume resources without creating sufficient economic value.

Research on product-level profitability supports the need for data-driven portfolio analysis. Product-level profitability research has emphasized the importance of connecting product structure, transaction data, business processes and information systems to fact-based product portfolio management. Other research has demonstrated that profitability can vary significantly across products and that activity-based costing can reveal difference that simpler costing approaches may not capture.

This project applies these ideas to a practical analytics case involving Nassau Candy Distributor. The central research question is:

How can product-level profitability and margin-risk analytics be used to identify priority management actions?

2.Literature Review

2.1 Product-Level Profitability

Product profitability is not uniform across a portfolio. Research by Hellstrom et al. emphasizes the challenges and preconditions for data-driven, fact-based product portfolio management, including the integration of product, transaction and business data. This supports the approach used in this project, where product-level sales, cost and profit are brought together in a common analytical view.

Dhar and Hoach examined causes and effects of product profitability and found significant variation in profitability across products. Their work also illustrates how more detailed costing can reveal differences between products that may otherwise be hidden. This is directly relevant to the dashboard's use of cost percentage of sales, gross margin and profit per unit as complementary measures.

2.2 Product Portfolio and Profitability

MacMillan, Hambrick and Day examined the relationship between product portfolio characteristics and profitability using PIMS-based evidence. Their findings support the proposition that product portfolio characteristics are associated with differences in business performance. In the present study, the portfolio is examined through contribution ranking, risk flags and concentration analysis.

2.3 Business Intelligence and Decision Support

Recent business intelligence literature describes dashboards as an important interface between analytical systems and managerial decisions making. A 2024 experimental study of dashboard visualizations found that the format, currency and completeness of information can influence decision quality indirectly by reducing perceived task complexity and increasing information satisfaction.

This finding supports the design objective of the Streamlit dashboard: rather than presenting only raw tables, the application combines headline KPI's, comparison, risk categories, Pareto charts and actionable recommendations. The goal is to reduce the effort required to identify which products deserve attention.

2.4 Profit Concentration and Risk

Pareto-style analysis is useful in management because it highlights concentration: a relatively small number of products can account for a large share of total profit. In this project, the dashboard explicitly calculates cumulative gross profit and reports the number of products required to reach the 80% cumulative-profit level.

The project also incorporates margin risk and cost-risk categories. This approach recognizes that a product can be strategically important because of its contribution while simultaneously presenting operational or cost-management risk.

3. Research Objectives

- Measure overall product-line profitability using revenue, cost, gross profit and gross margin.
- Compare profitability across products and divisions.
- Identify products with high cost exposure and low margins.
- Measure each product's contribution to total gross profit.
- Identify concentration of profit using Pareto analysis.
- Evaluate margin volatility over time.
- Translate analytical results into management recommendations.
- Demonstrate how an interactive Streamlit dashboard can support data-driven decision making.

4. Methodology

The project follows a descriptive and diagnostic business analytics methodology. The underlying workflow uses Python and pandas for data preparation and calculation and Streamlit for interactive visualization. The dashboard shown in the supplied evidence automatically loads the dataset from the Downloads location.

4.1 Analytical Measures

Measure	Definition / Purpose
Revenue	Total sales generated by the selected portfolio.
Cost	Cost associated with the selected sales.
Gross Profit	Revenue less cost.
Gross Margin (%)	Gross profit divided by sales.

Profit per Unit	Gross profit relative to units.
Revenue Contribution	Product sales divided by total sales.
Profit Contribution	Product gross profit divided by total gross profit.
Cost % of Sales	Cost divided by sales.
Margin Volatility	Variation in gross margin over time.

4.2 Dashboard Controls

The documented dashboard contains a date-range selector covering 2024/01/02 to 2025/12/31 in the displayed configuration, division filters for Chocolate, Other and Sugar, a margin-threshold slider set to 30%, and a product-search field. These controls enable users to move from the portfolio view to targeted product or division investigations.

5. Results and Analysis

5.1 Portfolio-Level Profitability

The main dashboard displays revenue of \$141,784 and gross profit of \$93,443. Gross margin is 65.9% while cost represents 34.1% of sales. Profit per unit is displayed as \$2.42. These values indicate that the portfolio generates a substantial gross-profit share relative to sales in the selected dashboard configuration.

Portfolio KPI	Dashboard Value
Revenue	\$141,784
Gross Profit	\$93,443
Gross Margin	65.9%
Profit per Unit	\$2.42
Cost % of Sales	34.1%

Margin Volatility	3.36 percentage points
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5.2 Product-Level Margin Performance

The product leaderboard shows wide variation in gross margin and cost exposure. The highest displayed margin among the listed products is 80% for Everlasting Gobstopper, followed by Hair Toffee at 77.8% and Wonka Bar-Nutty Crunch Surprise at 71.3%. The lower end of the portfolio includes Wonka Gum at 52%, Laffy Taffy at 62.3% and Fizzy Lifting Drinks at 60%.

Product	Sales	Cost	Gross Profit	Gross Margin	Cost % Sales	Risk
Everlasting Gobstopper	\$130	\$26	\$104	80%	20%	Healthy
Hair Toffee	\$76	\$17	\$60	77.8%	22.2%	Healthy
Wonka Bar- Nutty Crunch Surprise	\$23,375	\$6,755	\$16,620	71.3%	28.7%	Healthy
Wonka Bar-Scrumdiddlyumptious	\$27,875	\$8,517	\$19,358	69.4%	30.6%	Healthy
Wonka Bar-Fudge Mallows	\$24,890	\$8,297	\$16,594	66.7%	33.3%	Healthy
Wonka Bar-Milk Chocolate	\$26,868	\$9,424	\$17,443	64.9%	35.1%	Healthy
Wonka Bar-Triple Dazzle Caramel	\$28,485	\$9,875	\$18,610	65.3%	34.7%	Healthy
Laffy Taffy	\$54	\$20	\$33	62.3%	37.7%	Healthy
Fizzy Lifting Drinks	\$79	\$32	\$47	60%	40%	Healthy

Wonka Gum	\$598	\$287	\$311	52%	48%	Healthy
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The leaderboard demonstrates why revenue should not be the only decision variable. For example, Wonka Bar-Scrumdiddlyumptioud has the largest displayed gross profit at \$19,358 among the products shown, while Everlasting Gobstopper has a much higher 80% gross margin but only \$104 gross profit because of its much smaller sales volume.

5.3 Division Performance

The Division Performance Summary shows major differences in scale. Chocolate generates \$131,693 in sales and \$88,825 in gross profit, with a 67.4% gross margin. Other generates \$9,663 in sales and \$4,333 in gross profit, with a 44.9% gross margin. Sugar generates \$247 in sales and \$283 in gross profit, with a 66.6% gross margin.

Division	Sales	Cost	Gross Profit	Gross Margin	Profit/Unit
Chocolate	\$131,693	\$42,868	\$88,825	67.4%	\$2.28
Other	\$9,663	\$5,330	\$4,333	44.9%	\$3.49
Sugar	\$427	\$143	\$283	66.6%	\$2.08

Chocolate is the dominant division in absolute financial contribution. However, Other has a materially lower gross margin (44.9%) and therefore represents a more immediate area for cost and pricing investigation despite its smaller sales base.

5.4 Cost vs Margin Diagnostics

The cost-versus-margin diagnostic plots cost percentage of sales against gross margin and overlays management thresholds. The dashboard uses a 50% cost threshold in the displayed division diagnostic and a 30% margin threshold in the dashboard controls. The purpose is to identify products that combine high cost exposure with weak margins.

5.5 Margin Risk Flags

The dashboard classifies the reviewed products into Healthy, Cost Risk and High Risk categories. The displayed charts show 11 Healthy products, 3 Cost Risk products and 1 High Risk products, for 15 products reviewed.

Risk Category	Products
Healthy	11
Cost Risk	3
High Risk	1
Total Reviewed	15

5.6 Pareto Profit Concentration

The Pareto analysis reports that approximately 4 of 15 products (26.7%) account for the first 80% of cumulative gross profit. This is an important management finding because a relatively small group of products drives a disproportionately large share of profit. The leading contributors displayed are Wonka Bar- Scrumdiddlyumptious (\$19,358), Wonka Bar- Triple Dazzle Caramel (\$18,610), Wonka Bar-Milk Chocolate(\$17,443) , Wonka Bar-Nutty Crunch Surprise (\$16,820) and Wonka Bar-Fudge Mallows (\$16,594).

5.7 Dependency Indicators

The dashboard reports four high-cost products, one low-margin product, one high-risk ,product and fifteen products reviewed. These indicators provide a compact management summary of portfolio vulnerabilities.

5.8 Priority Actions

The management recommendation layer converts analytical results into three action groups: Discontinuation Review, Cost Renegotiation and Monitor. The displayed counts are one product for Discontinuation Review, three for Cost Renegotiation and eleven for Monitor.

Product	Sales	Cost	Gross Profit	Margin	Risk	Recomendation
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Kazookles	\$1,206	\$1,113	\$93	7.7%	High Risk	Discontinuation Review
Fun Dip	\$12	\$7	\$5	40%	Cost Risk	Cost Renegotiation
Nerds	\$15	\$8	\$7	46.7%	Cost Risk	Cost Renegotiation
SweetARTS	\$62	\$33	\$29	46.7%	Cost Risk	Cost Renegotiation
Lickable Wallpaper	\$7,860	\$3,930	\$3,930	50%	Healthy	Monitor

Kazookles is the strongest immediate exception. Its 7.7% gross margin and 92.3% cost share of sales are substantially weaker than the broader portfolio. The dashboard therefore recommends a Discontinuation Review rather than routine monitoring,

5.9 Margin Volatility

The dashboard reports margin volatility of 3.36 percentage points in the displayed configuration. The time-series chart shows repeated short-term movements and several sharp downward spikes in gross margin. This suggests that aggregate margin stability can conceal individual periods of significant variation. Management should investigate the dates associated with the largest drops and determine whether they reflect pricing, product mix, cost changes, order-level effects or data-quality issues.

6. Discussion

The results demonstrate three complementary dimensions of profitability. First, absolute contribution matters: products such as Wonka Bar-Scrumdiddlyumptious generate substantial gross profit and therefore deserve protection. Second, margin efficiency matters: products with low gross margins and high cost percentages may require pricing or sourcing intervention even when their sales are modest. Third, Concentration and volatility matter: reliance on a small group of products and variation in margin over time create additional management risks.

The findings are consistent with prior research emphasizing product-level profitability analysis and data-driven portfolio management. They also illustrate why an interactive dashboard can be useful: the same portfolio can be viewed by division, product, margin threshold or risk category, allowing the user to move from descriptive performance to diagnostic investigation and action.

7. Business Recommendations

1. Conduct an immediate review of Kazookles. Validate pricing, sourcing, logistics and activity costs. If the low margin cannot be corrected economically, evaluate product discontinuation.

2. Renegotiate costs for Fun Dip, Nerds and SweetARTS. These products are classified as Cost Risk. Supplier terms, procurement quantities, packaging, freight and other controllable cost drivers should be reviewed.

3. Protect the leading profit contributors. The four products that reach approximately 80% cumulative profit should receive strong availability, pricing and supply-chain monitoring because disruption could materially affect total profit.

4. Investigate the Other division. Its 44.9% gross margin is substantially below Chocolate and Sugar in the displayed results. A cost and pricing review could identify opportunities for improvement.

5. Monitor margin volatility. Investigate the largest downwards spikes in the margin-over-time chart and establish recurring alerts for abnormal movements.

6. Maintain the Streamlit dashboard as a management tool. Use the dashboard regularly rather than as a one-time project deliverable. Track changes in risk flags, profit concentration and division performance.

8. Conclusion

This study demonstrates how product-level profitability analytics can convert transactional data into actionable management intelligence. The Nassau Candy Distributor dashboard shows a portfolio with a strong aggregate gross margin of 65.9%, but the portfolio is not uniformly profitable. Product-level margins, cost exposure and profit contributions vary considerably, and the portfolio exhibits meaningful profit concentration.

The most important management insight is that aggregate profitability should not be treated as sufficient evidence of portfolio health. Kazookles illustrates the opposite case: a product can remain part of the portfolio while generating only a 7.7% margin and absorbing 92.3% of sales in cost. At the same time, a small group of high-contribution products generates the majority of cumulative gross profit and therefore deserves protection.

The Streamlit implementation provides a practical bridge between analysis and decision making by combining KPIs, comparisons, diagnostics, risk classification, Pareto analysis, dependency indicators and management recommendations. Future work should connect the dashboard to more granular cost drivers and formalize recurring management reviews so that analytical insights result in measurable operational actions.

9. Limitations

The quantitative results in this paper are based on the dashboard views and project information supplied in the conversation. The paper therefore reports the values visible in those views rather than independently recomputing every metric from the raw CSV.

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Appendix A - Streamlit Dashboard Evidence

The following screenshots were supplied as project evidence. In the final submission

Figure A1.Overall Product Profitability Overview

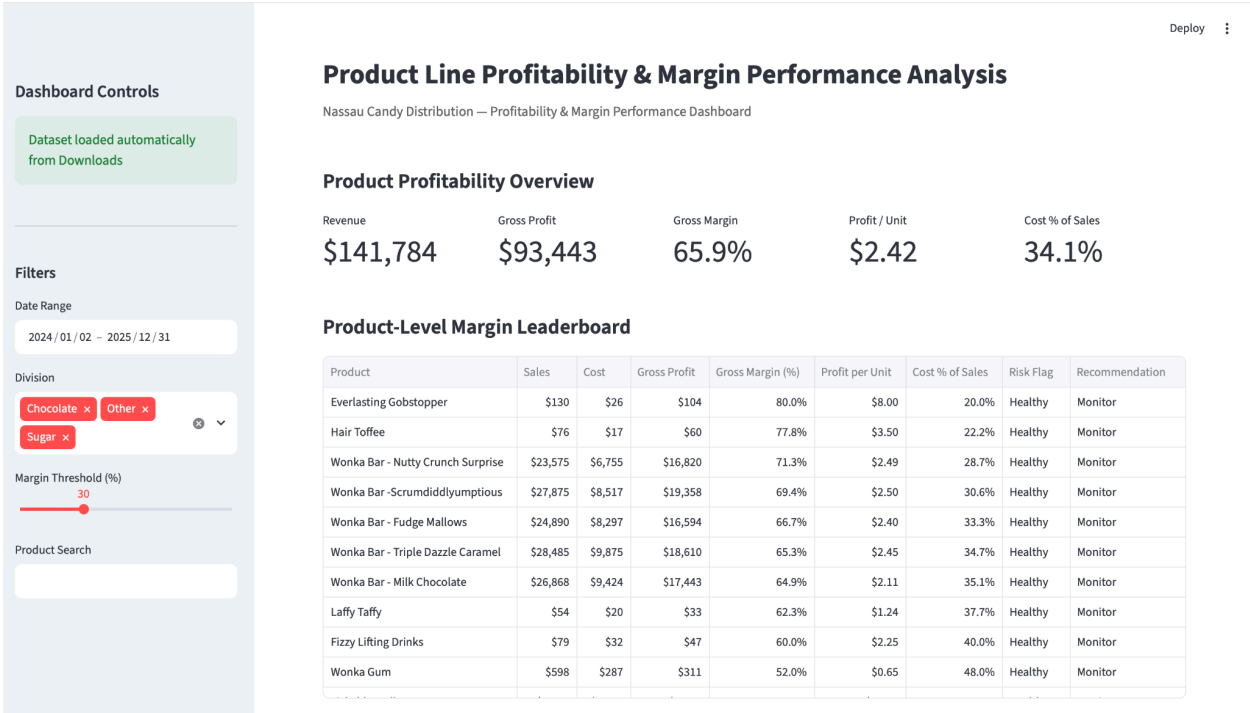


Figure A2. Profit Contribution / Top Products

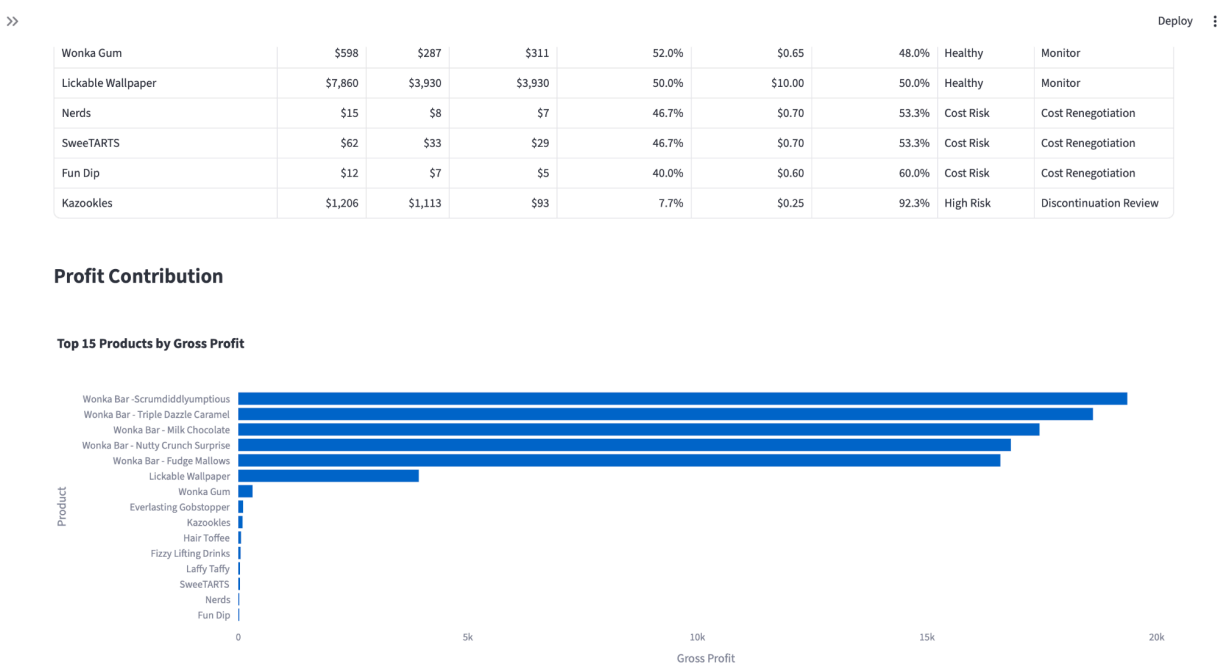


Figure A3. Division Performance

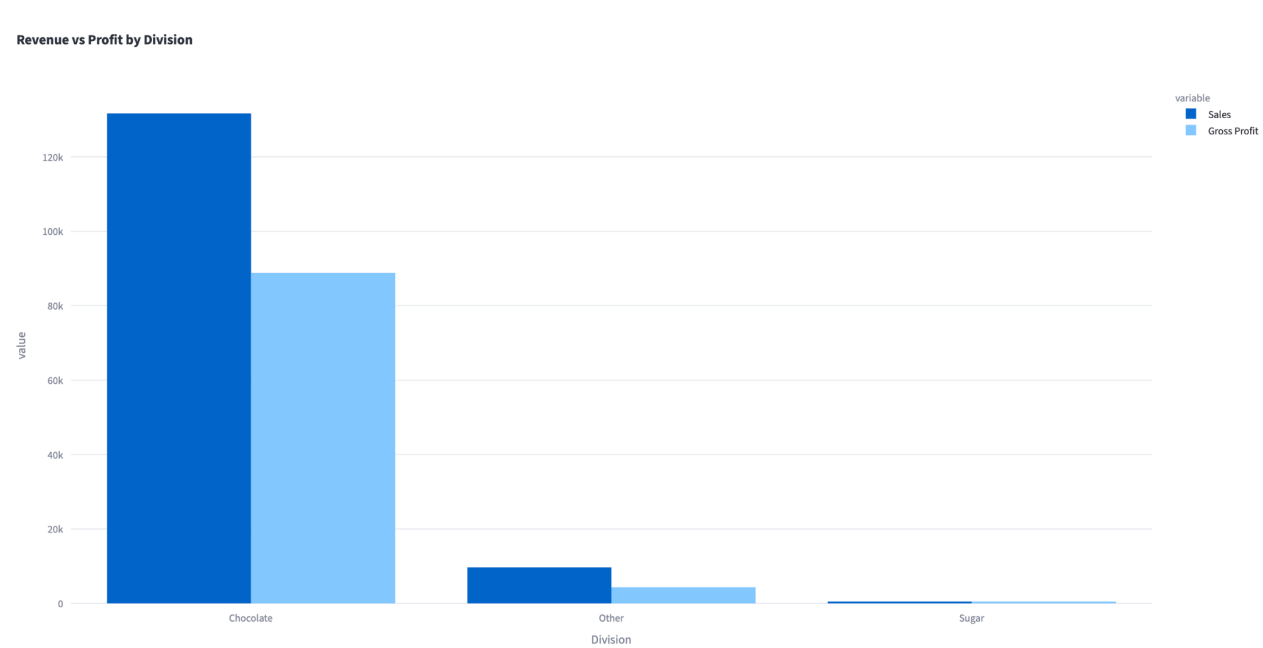


Figure A4 .Cost-vs-Margin Diagnostics

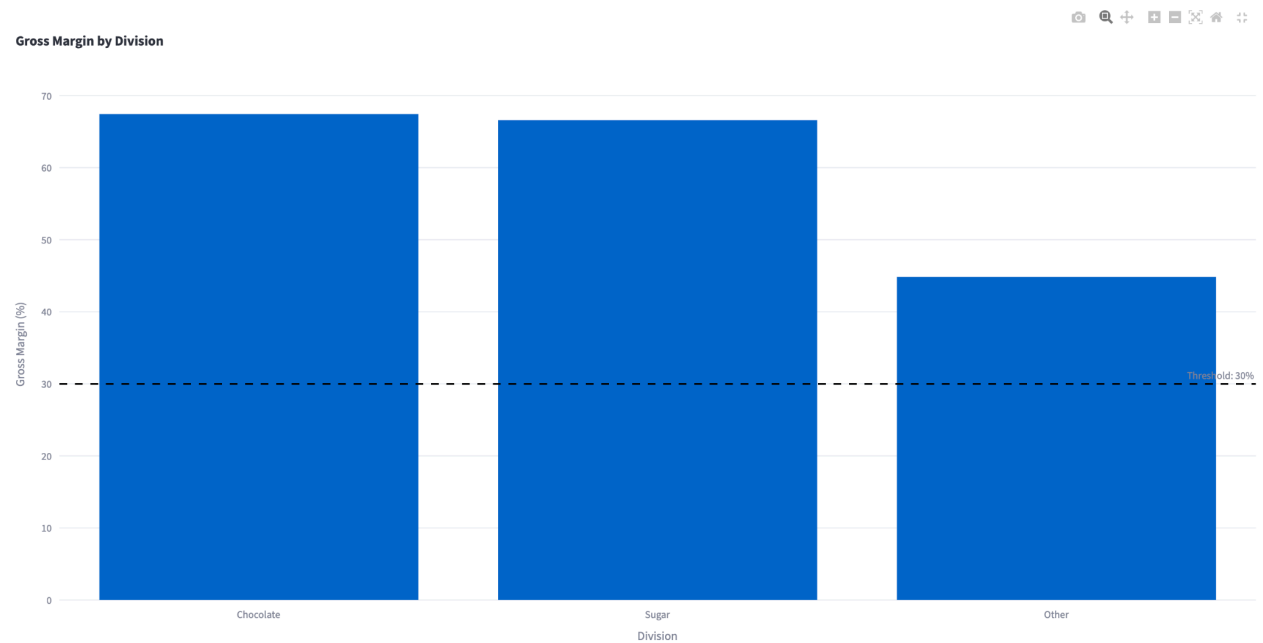


Figure A5.Division-Level Margin Risk Analysis

Division Performance Summary

Division	Sales	Cost	Gross Profit	Gross Margin (%)	Profit per Unit
Chocolate	\$131,693	\$42,868	\$88,825	67.4%	\$2.38
Other	\$9,663	\$5,330	\$4,333	44.8%	\$3.49
Sugar	\$427	\$143	\$285	66.6%	\$2.08

Cost vs Margin Diagnostics

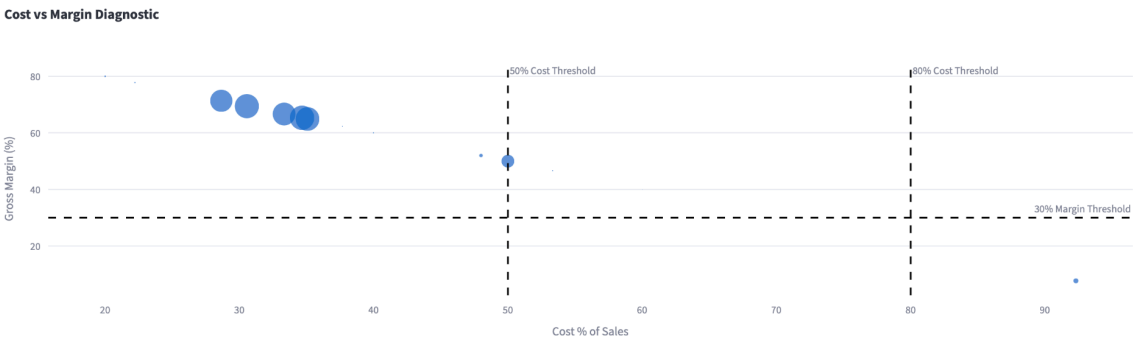


Figure A6. Product-Level Pareto Concentration Analysis

Products by Risk Category

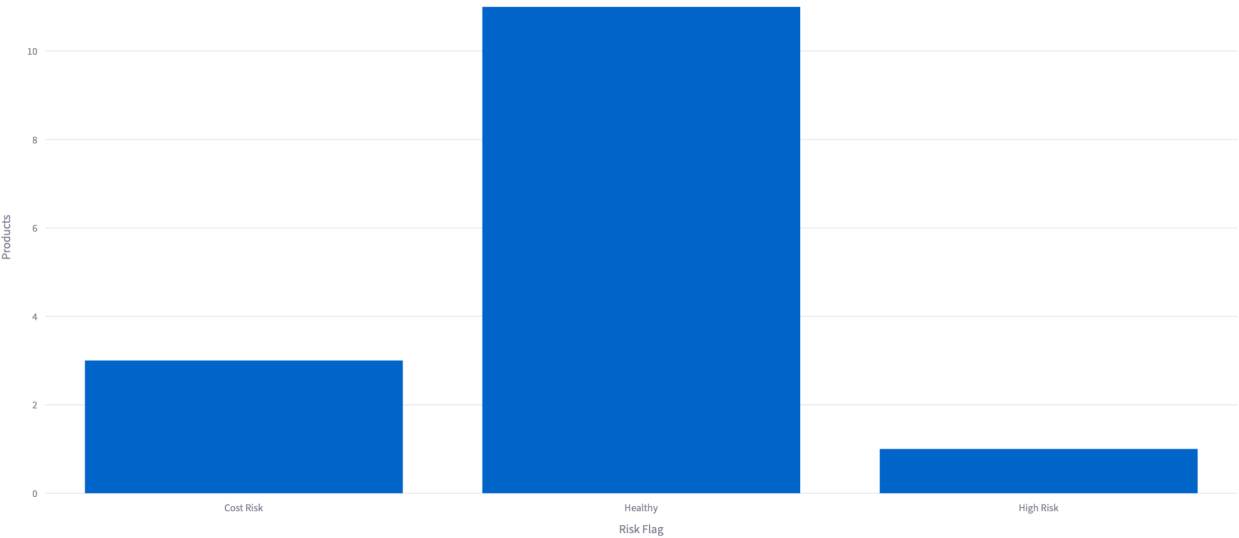
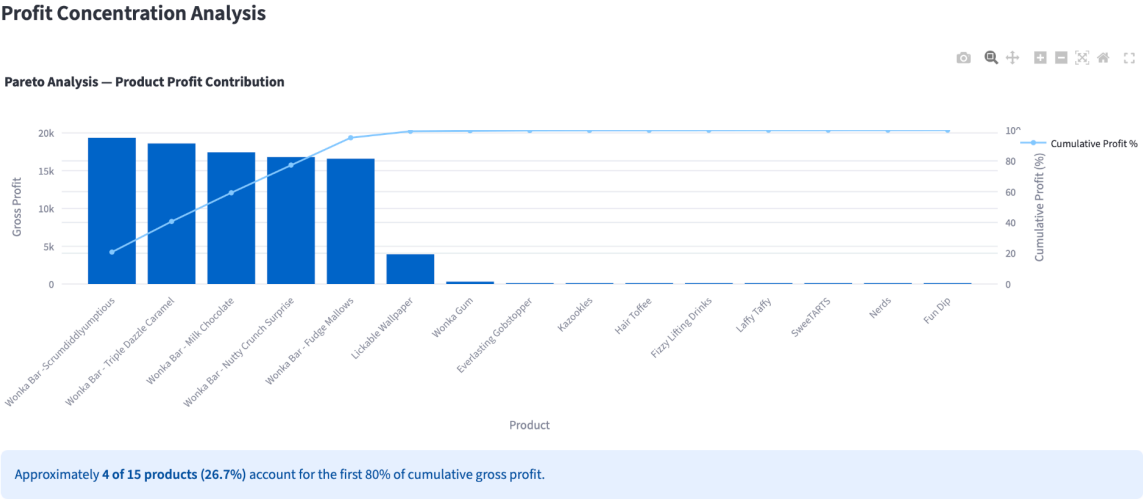


Figure A7. Pareto Analysis of Product-Level Profit Contribution



Dependency Indicators



Figure A8. Management Recommendation

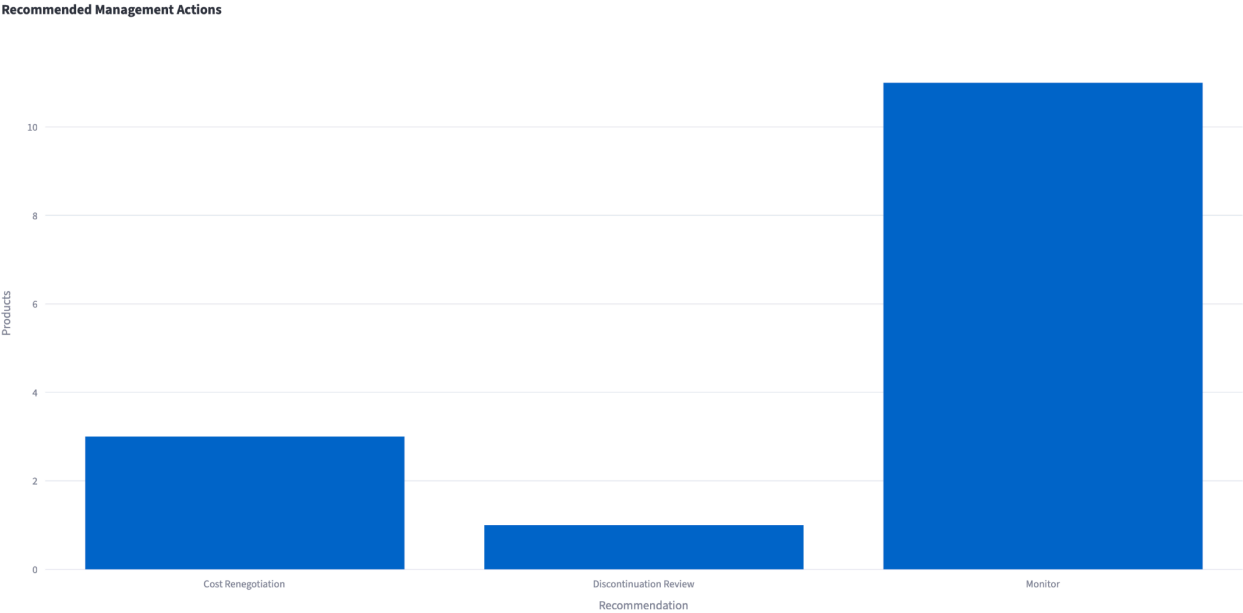


Figure A9.Product Action List and Priority Actions

Product Action List

Product	Gross Margin (%)	Cost % of Sales	Gross Profit	Profit per Unit	Risk Flag	Recommendation
Fun Dip	40.0%	60.0%	\$5	\$0.60	Cost Risk	Cost Renegotiation
Nerds	46.7%	53.3%	\$7	\$0.70	Cost Risk	Cost Renegotiation
SweetARTS	46.7%	53.3%	\$29	\$0.70	Cost Risk	Cost Renegotiation
Kazookles	7.7%	92.3%	\$93	\$0.25	High Risk	Discontinuation Review
Lickable Wallpaper	50.0%	50.0%	\$3,930	\$10.00	Healthy	Monitor
Wonka Gum	52.0%	48.0%	\$311	\$0.65	Healthy	Monitor
Fizzy Lifting Drinks	60.0%	40.0%	\$47	\$2.25	Healthy	Monitor
Laffy Taffy	62.3%	37.7%	\$33	\$1.24	Healthy	Monitor
Wonka Bar - Milk Chocolate	64.9%	35.1%	\$17,443	\$2.11	Healthy	Monitor
Wonka Bar - Triple Dazzle Caramel	65.3%	34.7%	\$18,610	\$2.45	Healthy	Monitor

Priority Actions

Discontinuation Review (1 products)

Product	Sales	Cost	Gross Profit	Gross Margin (%)	Cost % of Sales	Risk Flag
Kazookles	\$1,206	\$1,113	\$93	7.7%	92.3%	High Risk

Cost Renegotiation (3 products)

Product	Sales	Cost	Gross Profit	Gross Margin (%)	Cost % of Sales	Risk Flag
Fun Dip	\$12	\$7	\$5	40.0%	60.0%	Cost Risk
Nerds	\$15	\$8	\$7	46.7%	53.3%	Cost Risk
SweetARTS	\$62	\$33	\$29	46.7%	53.3%	Cost Risk

Monitor (11 products)

Product	Sales	Cost	Gross Profit	Gross Margin (%)	Cost % of Sales	Risk Flag
Everlasting Gobstopper	\$130	\$26	\$104	80.0%	20.0%	Healthy
Fizzy Lifting Drinks	\$79	\$32	\$47	60.0%	40.0%	Healthy
Hair Toffee	\$76	\$17	\$60	77.8%	22.2%	Healthy
Laffy Taffy	\$54	\$20	\$33	62.3%	37.7%	Healthy
Lickable Wallpaper	\$7,860	\$3,930	\$3,930	50.0%	50.0%	Healthy
Wonka Bar - Fudge Mallows	\$24,890	\$8,297	\$16,594	66.7%	33.3%	Healthy
Wonka Bar - Milk Chocolate	\$26,868	\$9,424	\$17,443	64.9%	35.1%	Healthy
Wonka Bar - Nutty Crunch Surprise	\$23,575	\$6,755	\$16,820	71.3%	28.7%	Healthy
Wonka Bar - Triple Dazzle Caramel	\$28,485	\$9,875	\$18,610	65.3%	34.7%	Healthy
Wonka Bar -Scrumdiddlyumptious	\$27,875	\$8,517	\$19,358	69.4%	30.6%	Healthy

Figure A10. Additional Product-Level Evidence

Margin Volatility

Gross Margin Over Time



Margin Volatility

3.36 percentage points

Data Quality & Dataset Information			
Rows	Products	Divisions	Columns
10,194	15	3	18